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Question	Answer	Marks
1	**Defining the problem**	
	R is the independent variable and *T* is the dependent variable, or vary *R* and measure *T*	**1**
	keep *C* constant	**1**
	Methods of data collection	
	labelled diagram or correct symbols of workable circuit including: • (d.c.) power supply correctly positioned • (neon) lamp correctly positioned do not accept ohmmeter in circuit	**1**
	circuit diagram to determine resistance of resistors e.g. using ammeter and voltmeter OR ohmmeter	**1**
	method to determine period or *T*, e.g. use a stopwatch / timer / oscilloscope do not accept counting the flashes in a specified time	**1**
	circuit diagram showing voltmeter(s) or oscilloscope(s) to determine V_i and V_F	**1**
	Method of Analysis	
	plots a graph of *T* against *R*	**1**
	$K = \frac{\text{gradient}}{C}$	**1**
	$V_L = V_i - (V_i - V_F)e^K = V_i - (V_i - V_F)e^{\frac{\text{gradient}}{C}}$	**1**

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1	**Additional detail including safety considerations**	**Max 6**
	switch off (high voltage) circuit (before changing the resistor) / wear insulating gloves to prevent electrocution / shock	**D1**
	resistance of resistors linked to diagram is V / I for ammeter / voltmeter method or gradient of appropriate graph or resistance from ohmmeter	**D2**
	input voltage or V_i is constant	**D3**
	repeat experiment for each value of R and average T	**D4**
	90 V (or larger) power supply do not accept a.c. or signal generator	**D5**
	for stopwatch method: time 10 or more flashes and divide by number of flashes for oscilloscope method: length of wave × timebase	**D6**
	record value of capacitance from the capacitor or method to determine capacitance	**D7**
	appropriate circuit to enable capacitance to be determined	**D8**
	relationship valid if a straight line passing through the origin is produced	**D9**
	method to obtain a measurable time period e.g. do a preliminary experiment to choose appropriate resistors, use large values of R or C	**D10**

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